

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 4: Linear Inequations (In One Variable)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Solution set and the replacement set

1. Find the solution set of $0 < -\frac{x}{3} < 2, x \in Z$. (a) $\{-5, -4, -3, -2, -1\}$ (b) $\{-6, -5, -4, -3, -2, -1\}$ (c) $\{-5, -4, -3, -2, -1, 0\}$ (d) $\{-6, -5, -4, -3, -2, -1, 0\}$ **ICSE 2025, 1 mark**

SOLUTION

Multiply through by -3 . Multiplying by a negative number reverses the signs: $0 > x > -6$.

The integers between -6 and 0 (both excluded) are $-5, -4, -3, -2, -1$.

Answer: (a)

2 Combining inequations

2. Solve $2x - \frac{5}{3} < \frac{3x}{5} + 10 \leq \frac{4x}{5} + 11, x \in R$. Write the solution set and show it on the number line. **ICSE 2025, 3 marks**

SOLUTION

Left part: $2x - \frac{3x}{5} < 10 + \frac{5}{3}$, so $\frac{7x}{5} < \frac{35}{3}$ and $x < \frac{25}{3}$.

Right part: $10 - 11 \leq \frac{4x}{5} - \frac{3x}{5}$, so $-1 \leq \frac{x}{5}$ and $x \geq -5$.

Solution set: $\left\{x : -5 \leq x < 8\frac{1}{3}, x \in R\right\}$



Solid dot at -5 (included), hollow circle at $8\frac{1}{3}$ (excluded)

3. Solve $-3 + x \leq \frac{7x}{2} + 2 < 8 + 2x$, where x is an integer. Write the solution set and show it on the number line. **ICSE 2024, 3 marks**

SOLUTION

Left part: $-3 - 2 \leq \frac{7x}{2} - x$, so $-5 \leq \frac{5x}{2}$ and $x \geq -2$.

Right part: $\frac{7x}{2} - 2x < 8 - 2$, so $\frac{3x}{2} < 6$ and $x < 4$.

Solution set: $\{-2, -1, 0, 1, 2, 3\}$



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